

Future Soldier

2030 INITIATIVE

DATUM

02 | 2009

03. Initiative Charter

KLASSIFIKATION: CONCEPT FRAMEWORK

The Future Soldier 2030 Initiative is a conceptual capability framework — **not established doctrine** — developed to identify what soldiers will require in future combat environments and to guide investment, stimulate technical dialogue, and evaluate trade-offs across the warfighter system.

SEE § 04 - DOMAINS

Umfang (Scope)	Verzeichnungsbogen (Details)	Status
Lead Org	NSRDEC leads development as RDECOM's Soldier System Integration Domain head. Additional input from contributing RDECs, PEO-Soldier, and designated Knowledge Centers.	[LEAD]
Intent	Not established doctrine. Guides investment priorities, stimulates dialogue, and evaluates technological trade-offs for the future warfighter.	[ADVISORY]
Outcome	Seven cross-cutting functional domains, 27 entity nodes, 31 documented relationships forming the integrated soldier-as-a-system architecture.	[VERIFIED]

ORIGINATOR (NSRDEC)

DOMAIN AUTHORITY

DATE / DATUM

KLASSIFIKATION

Unclassified • For Reference

FORM

FS-2030 / Charter

ORIGINATOR

US Army NSRDEC

DOMAIN LEAD

Soldier System Integration

DATUM

02 • 2009

SOURCE

d2409ad840e3.pdf

REVISION

Rev. 1.0

CHARTER SUPERSEDED BY FY+10 REVIEW

SERIAL NSRDEC • FS2030 • 03 • CHARTER



04. Capability Domains

KLASSIFIKATION: SEVEN FUNCTIONAL AREAS

Domain	Capability Synopsis	Clearance
HPT	PIA paired with each warfighter; immersive VR/AR training; embedded physiological sensors; nootropic and prosthetic enhancement.	UNRESTRICTED
PROT	Multifunctional "second-skin" ensemble. Ballistic, blast, reactive flame, CBRN. Autonomous trauma care & microclimate control.	CROSS-REF FIGS TO § 05 UNRESTRICTED
LETH	Lightweight low-recoil platform firing caseless kinetic and air-bursting HE munitions. Sighting integrated to headgear.	UNRESTRICTED
MOB	Modular LBU exoskeleton + Mission Enhancement Chassis (MAC) for urban search-and-rescue, breaching, and load carriage.	UNRESTRICTED
NET	Cognitive networks & software-defined radio. Secure multi-band transmission through distributed conformal nano-antennas.	UNRESTRICTED
SENS	180° visor display, 360° hyper-spectral fusion, biometric ID, 3D audio queuing, autonomous swarming robotics.	UNRESTRICTED
PWR	Hybrid multi-fuel generator + polymeric nano-fiber batteries + magnetic-resonance wireless transfer.	UNRESTRICTED

CLASSIFICATION STAMP LIBRARY

CLASSIFIED

APPROVED

FS-2030
NSRDEC

ARCHIVED

FILED • §04

DOM
X7

FOR REFERENCE

VERIFIED

CARD CATALOGUE • DOMAIN INDEX

INDEX • § 04 • CAPABILITY DOMAINS

HPT • Human Performance & Training p. 04-A

PROT • Soldier Protection p. 04-B

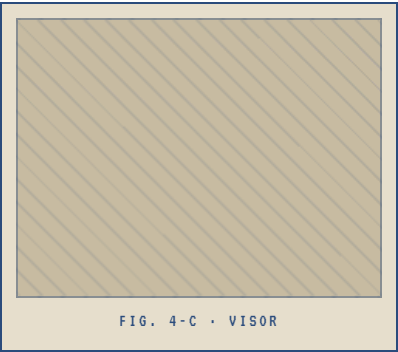
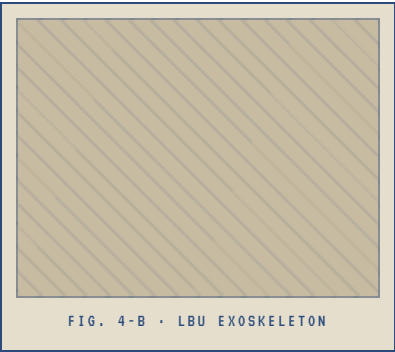
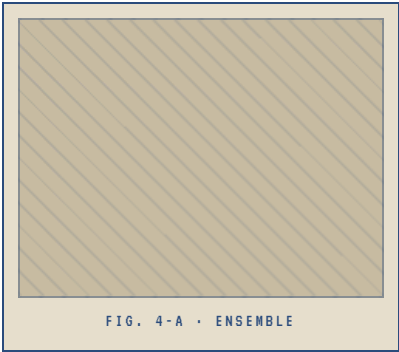
LETH • Lethality p. 04-C

MOB • Mobility & Logistics p. 04-D

NET • Soldier Network p. 04-E

SENS • Soldier Sensors p. 04-F

PWR • Soldier Power & Energy p. 04-G



TECHNICAL REVIEWER

DOMAIN AUTHORITY

DATE / DATUM

SECTION

§ 04 • Capability Domains

DOMAIN COUNT

07

LEAD TECH (HPT)

Personal Intelligent Agent

LEAD TECH (PROT)

Multifunctional Ensemble

LEAD TECH (MOB)

LBU Exoskeleton + MAC

CRITICAL RISK

Power & energy density

TRL RANGE

3 - 6

VERIFY TRL BEFORE CONTRACT

SERIAL NSRDEC • FS2030 • 04 • DOMAINS



05. Integration Matrix

KLASSIFIKATION:

CROSS-DOMAIN COUPLINGS

Edge	Coupling	Status
Visor ↔ Rifle	Sighting integrated to headgear display – virtual trigger and forward-observer capability tied to caseless kinetic rifle.	[COUPLED]
Ensemble ↔ Net	Distributed conformal nano-antenna arrays embedded into the protective ensemble; supports cognitive net & multi-band TX.	[COUPLED]
LBU ↔ Power	Exoskeleton augmentation drawn from central hybrid generator via magnetic-resonance wireless transfer.	[COUPLED]
PIA ↔ Warfighter	Personal Intelligent Agent monitors physiological state and cognitive load; mediates data presented to the warfighter.	[COUPLED]

SELECTED GRAPH EDGES

LEADS DEVELOPMENT OF	Organization: NSRDEC → Concept: Future Soldier 2030
CONTRIBUTES TO	Organization: RDECOM → Concept: Future Soldier 2030
CONTRIBUTES TO	Organization: PEO-Soldier → Concept: Future Soldier 2030
DEFINES CORE AREA	Concept: FS2030 → Domain: HPT · Protection · Lethality · Mobility · Network · Sensors · Power
UTILIZES	Domain: HPT → Technology: PIA · VR/AR
MONITORS STATE OF	Technology: PIA → Entity: Warfighter
IMPLEMENTED VIA	Domain: Protection → System: Multifunctional Ensemble
PROVIDES INHERENT	System: Ensemble → System: Reactive CBRN & Flame Barrier
INTERLOCKS WITH	System: LBU → System: Mission Enhancement Chassis
TRANSMITS THROUGH	System: Cognitive Nets → System: Conformal Nano-Antennas

DISPLAYSAINPOINTFOR

System: Visor → Weapon: Caseless Kinetic Rifle

EMBEDS

System: Ensemble → System: Nano-Antennas

POWEREDBY

System: LBU → System: Hybrid Generator

INTEGRATION AUTH.

DOMAIN AUTHORITY

DATE / DATUM

SECTION

§ 05 • Integration Matrix

NODE COUNT

27

EDGE COUNT

31

GRAPH FORMAT

Mermaid • JSON

CRITICAL COUPLING

Visor → Caseless Rifle

POWER BACKBONE

LBU ← Hybrid Generator

ANNEX A - FULL EDGE LIST

SERIAL NSRDEC • FS2030 • 05 • MATRIX

